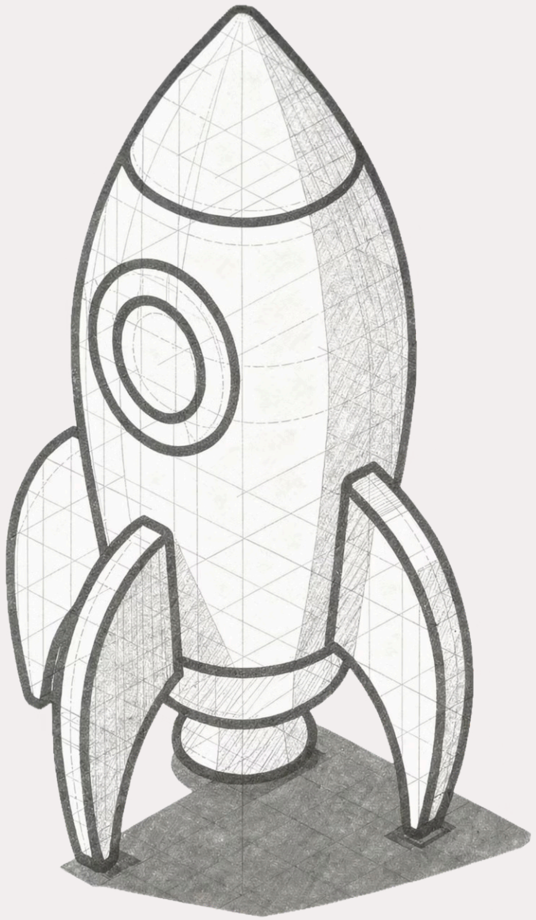
Day 8: Remote Execution Agents & Hybrid Execution in Astronomer Airflow

A deep dive into decoupled orchestration, secure agent-based execution, and hybrid cloud strategies for modern data pipelines.

MODULE 1 · REMOTE EXECUTION ARCHITECTURE



Module 1

Remote Execution Architecture

In this module, we explore how Astronomer Airflow separates orchestration from execution — enabling secure, compliant, and flexible data pipeline management across any environment.

1. **Hybrid Orchestration**
2. **Remote Execution Agents**
3. **On-Prem + Cloud**
4. **Security Model**

The Challenge: Orchestration Meets Data Sovereignty

Traditional Airflow architectures are centralized — orchestration and execution are tightly coupled. As organizations scale, this creates serious friction with modern compliance and security requirements.

Centralized Risk

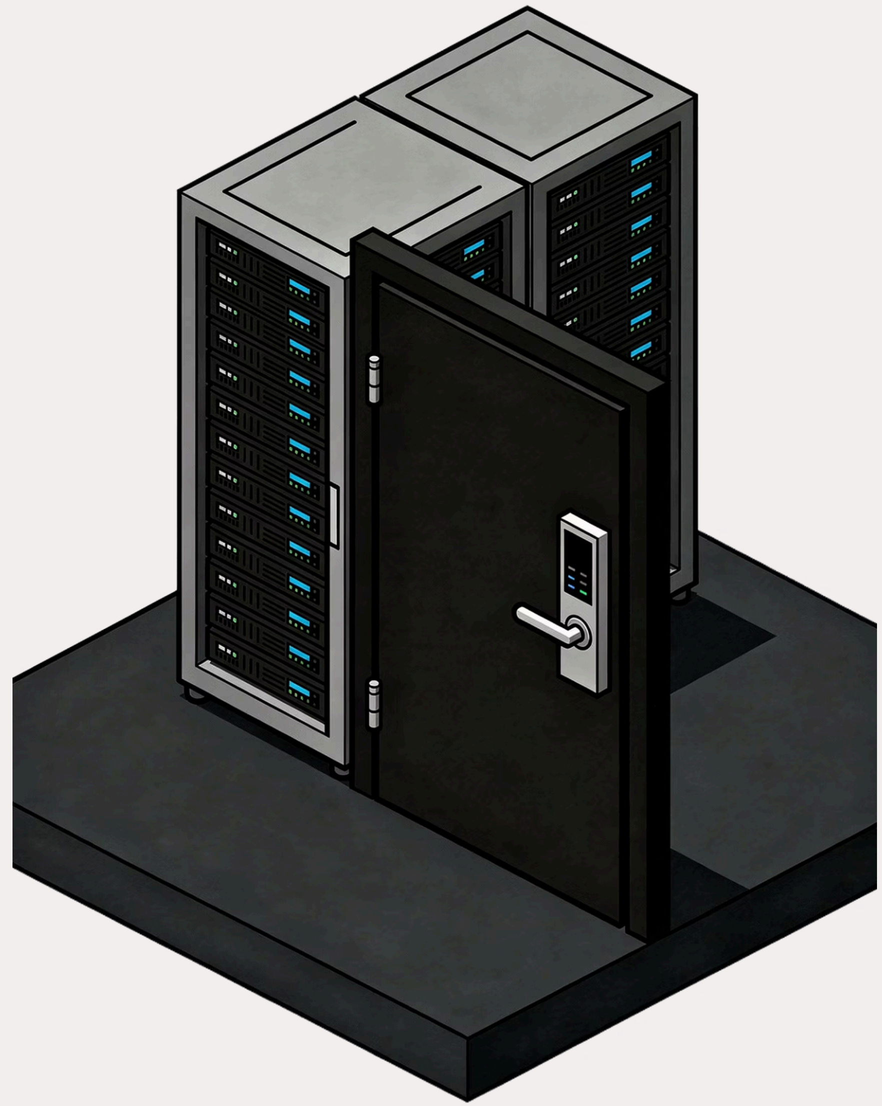
When orchestration and execution share the same plane, sensitive data is exposed to unnecessary movement and broader attack surfaces.

Compliance Pressure

Regulations like GDPR, HIPAA, and SOC 2 require strict data locality — data cannot freely traverse network boundaries.

Zero-Trust Gaps

Inbound network connections from cloud schedulers into private environments violate zero-trust principles and create vulnerabilities.



Introducing Remote Execution: Decoupling Orchestration and Execution

Remote Execution solves the sovereignty challenge by cleanly separating **where you orchestrate** from **where you execute**. The two planes operate independently, connected only through secure outbound channels.

Orchestration Plane

Astro-Managed

- Airflow Scheduler
- Web & API Server
- Metadata Database
- Remote Execution API

Hosted and managed entirely by Astronomer

Execution Plane

Customer-Controlled

- Your on-prem or cloud Kubernetes cluster
- Remote Execution Agents
- Sensitive data and code stay local
- No inbound connections required

Sensitive data **never** leaves your execution plane

Remote Execution Agents: The Core of the Execution Plane

Agents are lightweight Kubernetes services deployed in your own infrastructure. They pull work from Astro using secure, outbound-only HTTPS connections — no open inbound ports required.

DAG Processor

Parses and serializes DAG code locally, ensuring logic stays within your environment.

Triggerer

Manages deferrable tasks asynchronously, freeing worker slots for active computation.

Worker

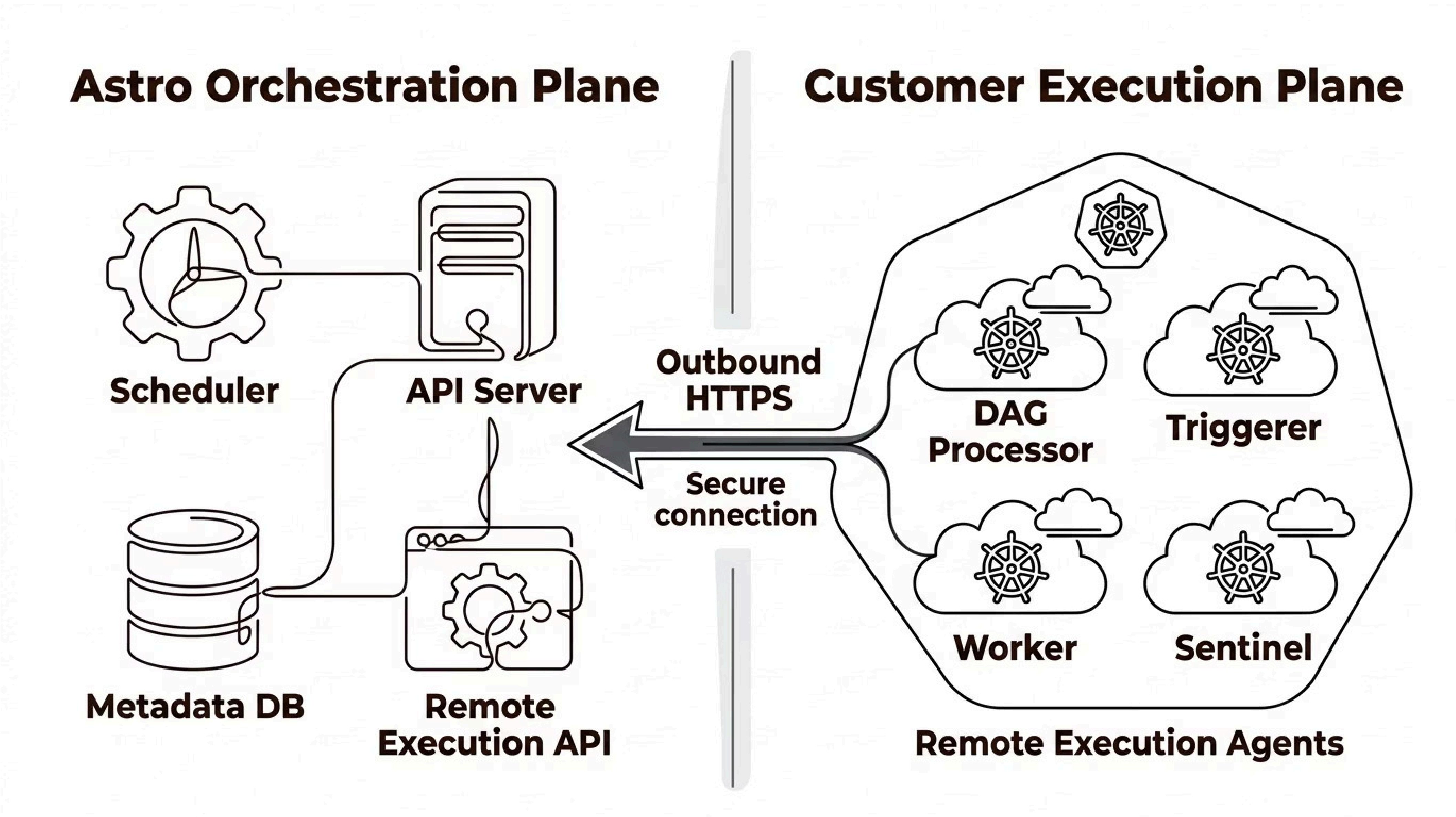
Executes Airflow tasks within your infrastructure, keeping data processing fully local.

Sentinel

Continuously monitors agent health and triggers rerouting if an agent becomes unavailable.

Architecture Overview: Orchestration to Execution

The diagram below illustrates the clean separation between Astro's managed orchestration plane and the customer-controlled execution plane, connected exclusively via secure outbound HTTPS.



How Agents Communicate: Heartbeats and Tokens

The communication model is built on a pull-based, outbound-only design. Agents reach out to Astro — Astro never reaches back into your network. This fundamentally eliminates inbound firewall exposure.

1

Agent Tokens

Securely authenticate each agent to the Astro orchestration plane. Tokens are generated in the Astro UI and scoped per deployment.

2

Outbound-Only Connections

All communication originates from the agent. No inbound traffic from Astro is ever required, preserving your firewall posture.

3

Heartbeat Mechanism

Agents report their capabilities and health status to the API Server on a regular interval, enabling real-time visibility.

4

Automatic Resilience

If an agent misses heartbeats and is declared unhealthy, the API Server automatically reroutes pending tasks to healthy agents.

Key Concepts: DAG Bundles

DAG Bundles are the packaging mechanism for delivering DAG code and supporting files to your execution agents. Choosing the right bundle type impacts versioning, deployment speed, and operational flexibility.

GitDagBundle *(Recommended)*

DAGs are stored in a Git repository. Changes are automatically versioned and tracked. This is the preferred approach for production environments, enabling GitOps workflows, code reviews, and rollback capabilities.

Best for: Production, CI/CD pipelines, team collaboration

LocalDagBundle

DAGs are baked directly into the container image or mounted via a persistent volume. Simpler to set up but requires image rebuilds or volume management for updates.

Best for: Development, testing, single-developer workflows

Security by Design: Keeping Data Local

Remote Execution isn't just a deployment pattern — it's a security architecture. Every design decision minimizes exposure and keeps sensitive data within boundaries you control.

Zero-Trust Architecture

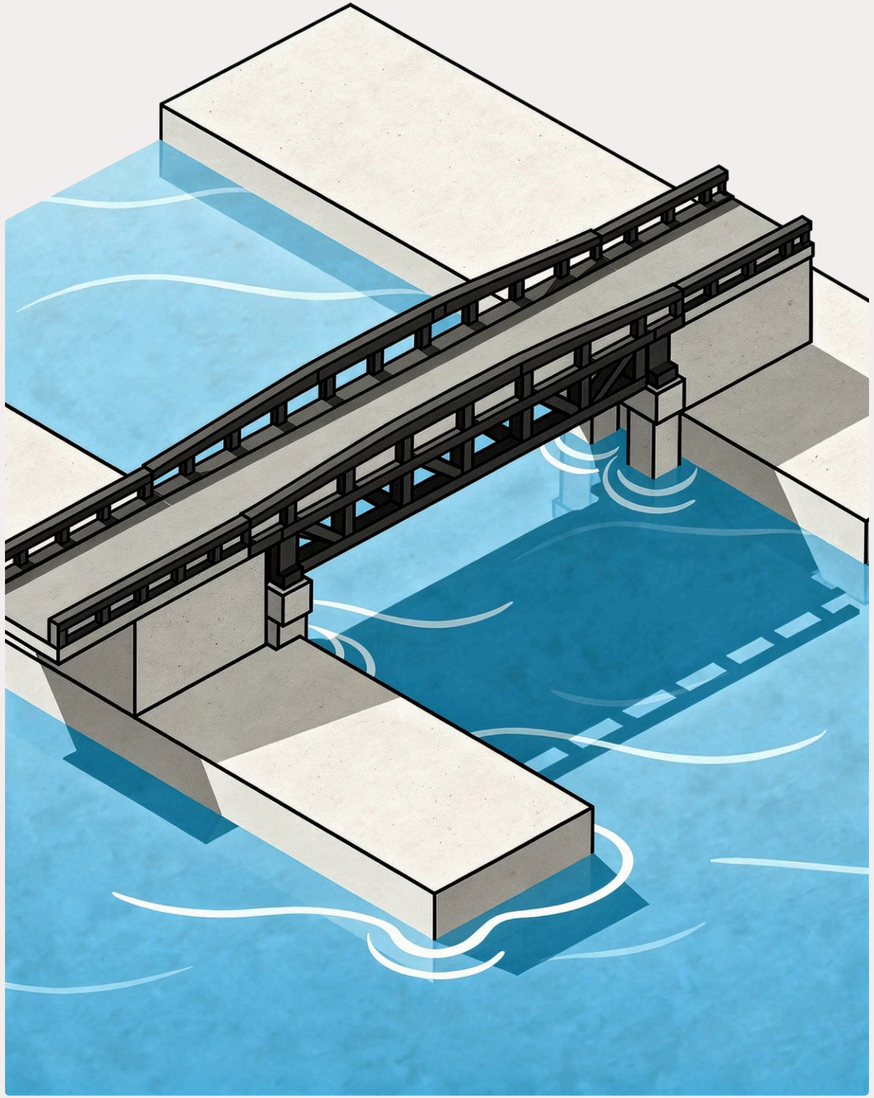
Eliminates inbound connections entirely. No open ports, no VPN tunnels. The attack surface is dramatically reduced because Astro never initiates connections into your network.

Data Sovereignty

Task execution, data processing, and sensitive logic remain inside your controlled perimeter. Your data never traverses to the Astro-managed cloud.

Regulatory Compliance

Designed to meet stringent standards including GDPR, HIPAA, SOC 2, and PCI-DSS — giving compliance teams the controls they require.



CHAPTER BREAK

From Remote to Hybrid

Now that we understand Remote Execution Agents, let's explore how they power **Hybrid Orchestration** — combining cloud-managed scheduling with on-premise execution for the best of both worlds.

Hybrid Orchestration: The Best of Both Worlds

Hybrid Orchestration is the strategic answer for organizations that need cloud agility without sacrificing control. It's not a compromise — it's an intentional architecture designed for complexity.

Cloud Agility

Leverage Astro's fully managed orchestration layer — no infrastructure to maintain, automatic scaling, and continuous updates — while retaining execution control.

On-Premise Control

Critical workloads, sensitive datasets, and proprietary models execute within your own infrastructure, satisfying data residency and sovereignty requirements.

Ideal Use Cases

Regulated industries (finance, healthcare, government), distributed teams, and organizations navigating complex multi-cloud or hybrid cloud strategies.

On-Prem + Cloud Orchestration: A Unified Workflow

In practice, hybrid orchestration means a single unified DAG can span cloud and on-premise resources — with Astro coordinating the sequence while your agents handle the sensitive execution.

The Scenario

Workflow orchestration is managed entirely by Astro in the cloud. The Scheduler determines task order and timing. Sensitive data processing — the actual task execution — happens on-premise, never leaving the private environment.

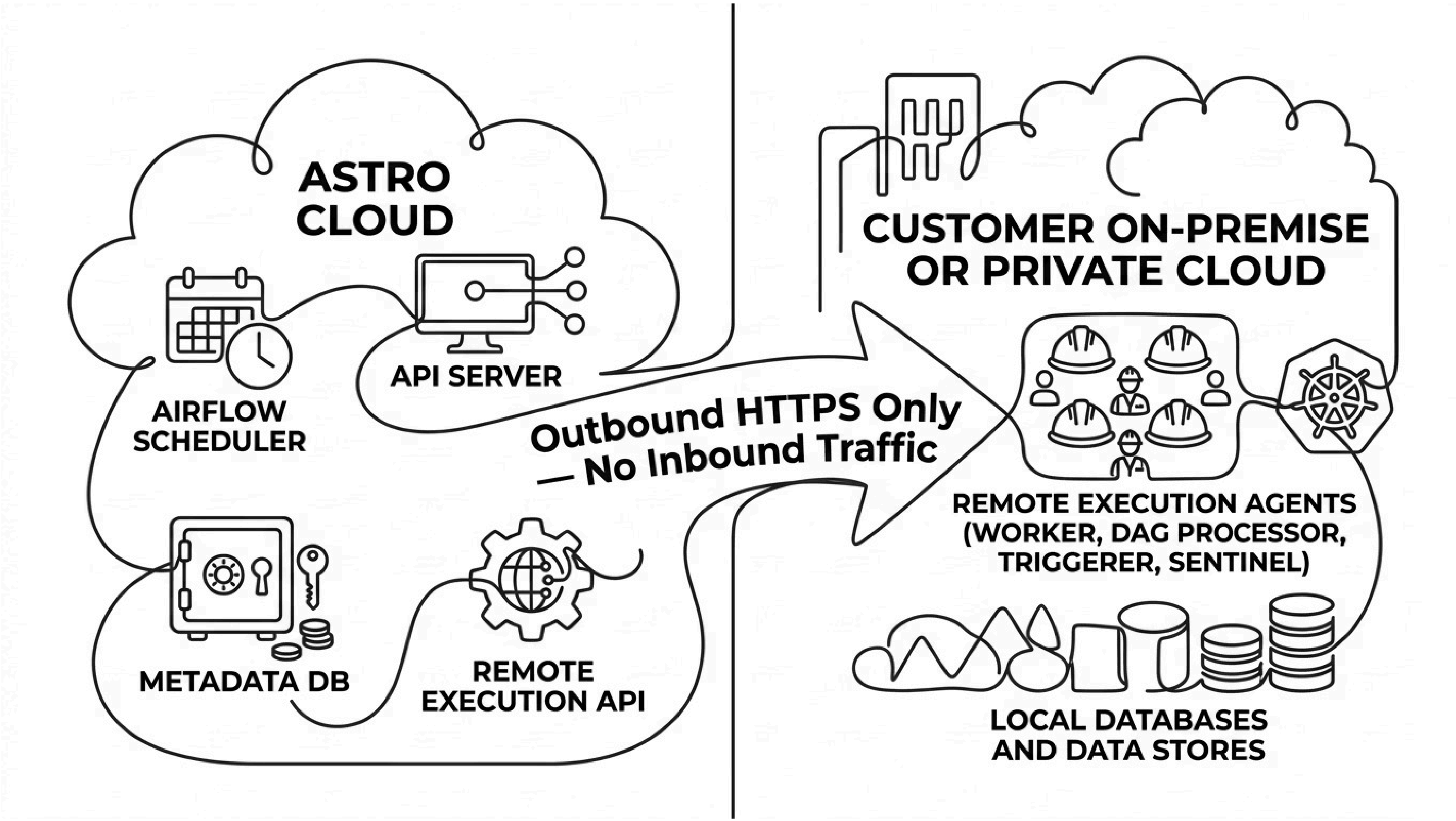
Real-World Example

A financial institution uses Astro to orchestrate its nightly settlement pipeline. The Scheduler triggers tasks on a precise schedule, but core transaction processing, fraud detection models, and customer data transformations all run on the bank's private on-premise Kubernetes cluster.

The cloud sees task metadata — never the transaction data itself.

Hybrid Architecture: Visual Reference

The following diagram captures a typical hybrid deployment — Astro Cloud handles scheduling and coordination, while the customer's private environment runs execution agents that process sensitive data locally.



Note: All arrows originate from the execution plane (right). The orchestration plane **never** initiates connections into the customer environment.

The Astro Executor: Powering Remote and Hybrid Modes

The Astro Executor is the default execution engine for Airflow 3.x deployments on Astronomer. It was purpose-built to manage the complexities of distributed, agent-based execution at scale.

Agent Lifecycle Management

Centrally manages agent registration, health tracking, and decommissioning across your deployment.

Intelligent Task Assignment

Routes tasks to the appropriate agents based on availability, affinity rules, and capability reporting.

Reliability & Performance

Delivers improved throughput, lower latency, and higher task success rates compared to traditional executors.

Cost Efficiency

Optimizes resource utilization across agents, reducing idle compute and infrastructure overhead.

Common Deployment Patterns

Astronomer supports three primary deployment patterns, giving organizations the flexibility to choose the right model for their security posture, infrastructure maturity, and operational preferences.

1

Hosted

Astro manages both orchestration **and** execution. The simplest model — ideal for teams without strict data residency requirements. Default for standard deployments.

2

Hybrid

Astro manages orchestration; tasks execute in your environment via Remote Agents. The recommended model for regulated industries and teams with sensitive workloads.

3

Fully Remote

All orchestration and execution occurs within your environment. Astro provides the control plane, but nothing runs in Astronomer's cloud. Maximum control and isolation.

Enterprise Tier Requirement

Remote Execution and Hybrid Orchestration are **Enterprise-tier features**. This tier is designed for organizations running demanding, mission-critical workloads that require the highest level of security, support, and capability.

Advanced Security

Enhanced access controls, audit logging, SSO/SAML integration, and dedicated security reviews.

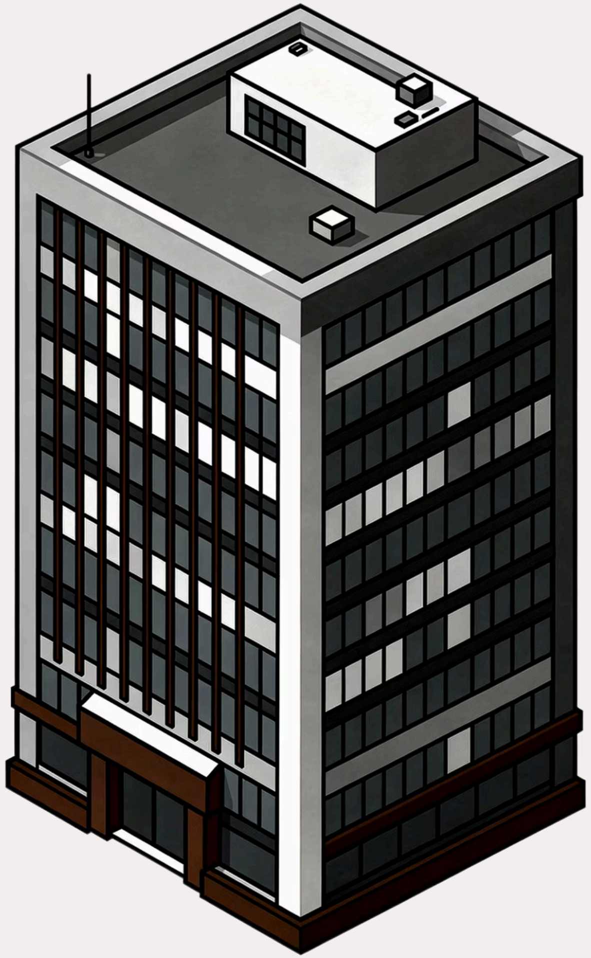
Dedicated Support

Priority SLAs, dedicated customer success management, and direct access to Astronomer engineering.

Advanced Capabilities

Full access to Remote Agents, hybrid deployment patterns, custom worker queues, and enterprise-grade observability.

⚠ Verify your Astronomer subscription tier before planning a Remote Execution deployment.



Step-by-Step: Registering a Remote Execution Agent

Getting your first Remote Execution Agent online takes just four steps. Each step is designed to be declarative and repeatable, making multi-agent deployments straightforward.

01

Create Agent Token

Navigate to the Astro UI → Deployments → Remote Agents. Generate a new secure token scoped to your target deployment. Store it safely — it will not be shown again.

02

Install Helm Chart

Add the Astronomer Helm repository and deploy the `remote-execution-agent` chart to your Kubernetes cluster, passing your Agent Token as a secret value.

03

Configure Allowed IPs

In the Astro UI, define the IP address ranges for your agent environment. This enforces network-level access control for outbound connections from your cluster.

04

Verify Heartbeat

Return to the Astro UI and confirm the agent appears with a healthy `CONNECTED` status. A green heartbeat indicator confirms successful registration and bidirectional communication.

Agent Registration in the Astro UI

Once your Helm deployment completes and the agent begins sending heartbeats, it appears in the Astro UI with real-time status indicators. You can monitor connectivity, view capability metadata, and inspect recent heartbeat timestamps — all from a single pane of glass.

A **green CONNECTED status** with a recent heartbeat timestamp confirms your agent is healthy and ready to receive tasks from the orchestration plane.

Summary: Secure, Flexible, Powerful Orchestration

Day 8 covered the architecture, components, and real-world patterns that make Astronomer Airflow a leading platform for enterprise-grade, distributed data pipelines.

Remote Execution Agents

Lightweight Kubernetes services that bring task execution into your environment — with zero inbound traffic and full data sovereignty.

Hybrid Orchestration

Cloud-managed scheduling paired with on-premise execution — ideal for regulated industries, distributed teams, and complex multi-cloud strategies.

Security by Design

Zero-trust architecture, outbound-only connections, and data locality controls that satisfy GDPR, HIPAA, SOC 2, and beyond.

Astronomer Platform

The Astro Executor, DAG Bundles, and Enterprise tier capabilities combine to deliver a robust, scalable foundation for modern data pipelines.

Next: Day 9 will cover advanced DAG authoring patterns and performance optimization in Airflow 3.x.